

UNIVERSITY COLLEGE LONDON
EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : COMP0089
MODULE NAME : Reinforcement Learning
LEVEL : Undergraduate (Masters Level)
DURATION : 3 Hours

MOCK EXAMINATION

Academic Year 2025–26

This paper contains **FOUR** mandatory questions.
Not all questions carry the same number of marks; see the mark allocation on each sub-question.

Read through **ALL** questions before attempting any.
Always provide justification and show intermediate work.
A correct but unsupported answer may not receive full marks.
Be concise and precise: do not include irrelevant information.
Standard calculators are allowed but are unlikely to be useful.

EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL.

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Useful Identities

Gaussian (normal) policy

$$\pi(a \mid s) = \frac{1}{\sqrt{2\pi} \sigma(s)} \exp\left(-\frac{(a - \mu(s))^2}{2 \sigma(s)^2}\right)$$

Softmax policy

$$\pi(a \mid s) = \frac{\exp(h(s, a))}{\sum_b \exp(h(s, b))}$$

Gradient of logarithm

$$\nabla \log f(x) = \frac{1}{f(x)} \nabla f(x)$$

Time step	Action	Reward
1	a_1	+1
2	a_2	+1
3	a_1	-1
4	a_3	+1
5	a_2	-1

Table 1: Observed bandit data.

Question 1A

Consider a multi-armed bandit with three arms: a_1, a_2, a_3 . The only possible rewards are $-1, 0$ and $+1$. We have been interacting with this bandit and so far have collected the data in Table 1.

Action values are maintained with the sample-mean rule: $Q_t(a) = \frac{1}{N_t(a)} \sum_{i: A_i=a} R_{i+1}$, where $N_t(a)$ is the number of times a was selected up to and including time t . If an action has never been selected, its value is 0. Ties are broken randomly throughout.

- (a) Suppose the policy is ε -greedy with $\varepsilon = 0.3$ with respect to the current action values. What is the policy at time $t = 6$? Give the policy in the form

$$\pi_6(a_1) = \dots, \quad \pi_6(a_2) = \dots, \quad \pi_6(a_3) = \dots$$

[3 marks]

- (b) What is the policy for the UCB algorithm with hyper-parameter $c = 1$ at time $t = 6$, after having seen the data in Table 1? Give the policy in the same form as above. [4 marks]
- (c) Suppose that the true expected reward of every arm is actually 0. We run UCB with $c = 1$ starting from the same data in Table 1. As we continue to interact with the bandit for $t \rightarrow \infty$, will the fraction of time spent on each arm converge? If so, to what fractions? [5 marks]

Question 1B

Consider an MDP with three states $\mathcal{S} = \{s_1, s_2, s_3\}$ and two actions $\mathcal{A} = \{a_L, a_R\}$. The transitions and rewards are deterministic:

State	Action	Next state	Reward
s_1	a_L	s_1	0
s_1	a_R	s_2	+2
s_2	a_L	s_1	+1
s_2	a_R	s_3	0
s_3	a_L	s_2	+1
s_3	a_R	s_3	0

The discount factor is $\gamma = 0.5$. Consider the uniformly random policy $\pi(a_L | s) = \pi(a_R | s) = 0.5$ for all s .

- (d) Compute the value function $v_\pi(s)$ for each state. [4 marks]
- (e) Consider linear function approximation with $\phi(s_1) = [1, 0]^\top$, $\phi(s_2) = [1, 1]^\top$, $\phi(s_3) = [0, 1]^\top$, so that $v_{\mathbf{w}}(s) = \mathbf{w}^\top \phi(s)$. Starting from $\mathbf{w} = [0, 0]^\top$ with learning rate $\alpha = 0.1$, and with the agent starting in s_2 at $t = 0$, compute the expected weight vector $\mathbb{E}[\mathbf{w}]$ after one step of semi-gradient TD(0) under the uniformly random policy. [5 marks]
- (f) Now change the features to $\phi(s_1) = [1]$, $\phi(s_2) = [2]$, $\phi(s_3) = [3]$ (a single scalar weight w). Suppose we run on-policy semi-gradient TD(0) under the uniformly random policy with a small constant learning rate α . Does this procedure converge as $t \rightarrow \infty$? If so, what is w at convergence? If not, does it diverge or oscillate? [6 marks]

[Total Q1: **27 marks**] [Running total: 27 marks]

Question 2

Consider a contextual bandit (single-state) setting with two actions $\{a_1, a_2\}$. The policy is parameterised by a single scalar $\theta \in \mathbb{R}$ via

$$\pi_\theta(a_1) = \sigma(\theta), \quad \pi_\theta(a_2) = 1 - \sigma(\theta),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the logistic sigmoid. We use the standard REINFORCE update *without* a baseline:

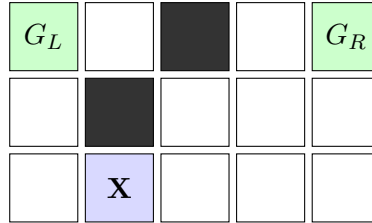
$$\theta \leftarrow \theta + \alpha R \frac{\partial}{\partial \theta} \log \pi_\theta(A).$$

- (a) Derive $\frac{\partial}{\partial \theta} \log \pi_\theta(a_1)$ and $\frac{\partial}{\partial \theta} \log \pi_\theta(a_2)$ in terms of $\sigma(\theta)$. [3 marks]
- (b) Suppose the current parameter is $\theta = 2$ and the learning rate is $\alpha = 0.5$. We select action $A = a_2$ and observe reward $R = +3$. What is θ after the update? [3 marks]
- (c) Now suppose $\theta = 0$ (so that $\pi_\theta(a_1) = \pi_\theta(a_2) = 0.5$). We again select $A = a_2$ and observe $R = +3$ with $\alpha = 0.5$. What is θ after the update? Compare the *magnitude* of the policy change (i.e. $|\pi_{\theta'}(a_1) - \pi_\theta(a_1)|$) to that from the previous sub-question. Which starting point leads to a larger change in the policy? Explain intuitively why. [5 marks]
- (d) Consider the expected update $\mathbb{E}_{A \sim \pi_\theta} [R(A) \frac{\partial}{\partial \theta} \log \pi_\theta(A)]$ for *general* reward functions $r(a_1)$ and $r(a_2)$. Show that this expected update is zero if and only if $r(a_1) = r(a_2)$. [5 marks]
- (e) Suppose $r(a_1) = +1$ and $r(a_2) = +5$. A colleague suggests adding a constant baseline $b = 3$ (subtracting b from the reward in the update). Another colleague suggests $b = 0$ (no baseline). For each choice, compute the variance of the stochastic gradient $\text{Var}_{A \sim \pi_\theta} [(R(A) - b) \frac{\partial}{\partial \theta} \log \pi_\theta(A)]$ as a function of θ . Evaluate both variances at $\theta = 0$ and determine which baseline gives lower variance. [7 marks]

[Total Q2: **23 marks**] [Running total: 50 marks]

Question 3

Consider the following grid MDP. The grid is 3×5 (3 rows, 5 columns). Black cells are walls. The agent starts at the cell marked **X**. The two exits are marked G_L (left) and G_R (right).



Actions are *up*, *down*, *left*, *right*. Attempting to move into a wall or off the grid results in staying in the current cell. Reaching either G_L or G_R terminates the episode; a new episode starts from **X**. The shortest path from **X** to G_L takes **3 steps**, and the shortest path from **X** to G_R takes **4 steps**.

Ties are broken randomly whenever relevant below.

- (a) Suppose all step rewards are -1 (including steps that bump into a wall). Exiting at G_L gives an additional reward of $+5$ (so the total reward on the exit step is $+5 - 1 = +4$), and exiting at G_R gives an additional reward of $+r$ (total on exit step: $r - 1$). The problem is undiscounted ($\gamma = 1$).

For which values of r is the optimal policy to head for G_R rather than G_L ? [3 marks]

- (b) Suppose $r = 50$ (so exiting right gives $+49$ on the exit step) and all other step rewards are -1 . We run a *uniformly random* policy π for many episodes and learn an undiscounted action-value function with SARSA until convergence (decreasing the learning rate appropriately). We then define a policy π' that is greedy with respect to the converged action values. What will π' be in the starting state **X**? [5 marks]

- (c) We now repeat the process: reinitialise all action values to zero and run π' from the previous sub-question for many episodes, again using SARSA until convergence. We construct a new greedy policy π'' . What is π'' in the starting state **X**? [4 marks]

- (d) Suppose instead we use Q-learning (off-policy) for both rounds of value learning, with the same uniformly random behaviour policy in *both* rounds. After the first round converges, we extract π'_Q . After the second round converges (still with uniform behaviour), we extract π''_Q . Give π'_Q and π''_Q in the starting state **X**. [5 marks]

- (e) Consider the original setup from part (b) with $r = 50$ and step cost -1 . Now add the following *reward shaping*: on every step, add a bonus $F(s, s') = \gamma \Phi(s') - \Phi(s)$ where $\Phi(s)$ equals the negative Manhattan distance from s to G_R and $\gamma = 0.95$.

If we run Q-learning to convergence under this shaped reward, is the greedy policy with respect to the converged values the same as the optimal policy under the original (unshaped) reward? Justify your answer. [7 marks]

[Total Q3: **24 marks**] [Running total: 74 marks]

Question 4

Consider a finite MDP with states $\mathcal{S}$, actions $\mathcal{A}$, transition probabilities $p(s' \mid s, a)$, expected rewards $r(s, a) = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$, and discount factor $\gamma \in [0, 1)$.

Define the following operator $\mathcal{T}$ on bounded action-value functions $q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$:

$$(\mathcal{T}q)(s, a) = r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) \frac{\max_b q(s', b) + \min_b q(s', b)}{2}.$$

That is, the operator bootstraps on the *average of the best and worst action values* in the next state, rather than just the max.

- (a) Is $\mathcal{T}$ a γ -contraction with respect to $\|\cdot\|_\infty$? If so, prove it. If not, give a counterexample. [5 marks]
- (b) Does $\mathcal{T}$ have a unique fixed point? Denote this fixed point by $\hat{q}_{\mathcal{T}}$. Is $\hat{q}_{\mathcal{T}}$ equal to the optimal action-value function q^* in general? If not, construct a concrete MDP (with explicit states, actions, transitions, and rewards) where they differ. [6 marks]
- (c) Consider applying $\mathcal{T}$ in a *deterministic* MDP (where each (s, a) leads to a unique next state). Does $\hat{q}_{\mathcal{T}} = q^*$ hold in this special case? If not, provide a counterexample. [4 marks]

For the following two sub-questions, consider a general finite MDP and two Markov policies π_1 and π_2 that are **not ordered**: there exist states s, s' such that $v_{\pi_1}(s) > v_{\pi_2}(s)$ and $v_{\pi_2}(s') > v_{\pi_1}(s')$.

- (d) Define the policy

$$\pi_{\max}(a \mid s) = \begin{cases} 1 & \text{if } a = \arg \max_b [q_{\pi_1}(s, b) + q_{\pi_2}(s, b)], \\ 0 & \text{otherwise} \end{cases}$$

(breaking ties arbitrarily). Is it true that $v_{\pi_{\max}}(s) \geq v_{\pi_1}(s)$ for all s ? Either prove it or construct a counterexample. [5 marks]

- (e) Define a different policy π' , based only on π_1, π_2 , and their value functions $(v_{\pi_1}, q_{\pi_1}, v_{\pi_2}, q_{\pi_2})$, that provably improves on *both* π_1 and π_2 simultaneously, i.e. $v_{\pi'}(s) \geq \max(v_{\pi_1}(s), v_{\pi_2}(s))$ for all $s \in \mathcal{S}$. Prove that your policy satisfies this property. [6 marks]

[Total Q4: 26 marks] [Total: 100 marks]

END OF PAPER